



Agile improvement practices in software organizations

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Abstract

Inspired by the chasm between early adopters and early majority in diffusion of technological innovations, this paper investigates how agile improvement practices can help software organizations successfully implement new processes. An action research-based improvement initiative implemented a new change-request process and tool at the telecom company Ericsson AB in Gothenburg, Sweden. The study identifies a 'guerilla tactic' that change agents can use in software organizations to cross the chasm, and it discusses lessons learned in relation to literature on diffusion of innovation and software agility. The contribution of the study is that the guerrilla tactic supports agile improvement practices and facilitates successful implementation.

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Introduction

Diffusion of innovations is both a practical challenge and an elaborated research topic (Zmud, 1984; Fichman & Kemerer, 1999; Ardis & Marcolin, 2001; Pries-Heje & Tryde, 2001; Moore, 2002; Rogers, 2003; Mustonen-Ollila & Lyytinen, 2004). Diffusion is a special type of communication in which the messages are about a new idea (Rogers, 2003, p. 6). An innovation is an idea, practice, or object that individuals or other units of adoption perceive as new (Rogers, 2003, p. 12). Rogers (2003) argues that a consequence of this perception is a degree of uncertainty, which implies a lack of predictability, structure, and information. To deal with this uncertainty and increase the likelihood of successful change, Paulk (1999) recommends using structured models of change and recognizing and addressing these models in the technology adoption curve (Rogers, 2003). One such model is 'the chasm' (Moore, 2002), which emphasizes the gap between early adopters and the early majority. Moore claims that failure to diffuse a new technology can stem from a failure to recognize and understand the challenges posed by the chasm. To cross the chasm, Moore (1999) introduces the 'down the bowling alley' tactic, that is, a way of working to successfully cross the chasm. A key feature of this tactic is that it addresses the uncertainties involved through agile improvement practices (Haeckel, 1995; Dove, 2001) in which the innovation is shaped and re-shaped through a series of interactions between change agents and adopters.

The literature on improving software organizations recognizes change agents and their tactics as important factors for successful change (McFeeley, 1996; Grady, 1997; Humphrey, 1989; Kautz *et al.*, 2001; Pries-Heje and Tryde, 2001; Rogers, 2003). It is therefore interesting to

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investigate how the chasm and the bowling alley tactic can inspire change agents in software organizations to develop agile improvement practices and thereby increase the likelihood of successful change. We studied and participated in an improvement initiative over a 16-month period at the telecom company Ericsson AB in Gothenburg, Sweden. The studied unit develops and maintains parts of 3G mobile networks and it has just over 500 employees. The unit's software improvement group consists of six people who are responsible for driving improvement initiatives. The initiative focused on defining and implementing a new change-request solution. Our research question was: *'How can change agents in software organizations adopt agile improvement practices to successfully cross the chasm?'* We organized the study as an action research effort (Susman and Evered, 1978) with two goals. First, we wanted to improve Ericsson's improvement practices inspired by Moore's ideas on how to successfully cross the chasm (Moore, 2002). Second, we wanted to contribute to the literature by developing knowledge on how to address uncertainty through agile improvement practices in software organizations.

To achieve these goals, the participating change agents developed and implemented a tactic that focused on establishing a trustworthy reference group and working closely with both the group as a whole and its individual members. They built this 'guerrilla tactic' on Moore's bowling alley approach (1999). Following DeLone & McLean's (1992) observation that information technology (IT) use is the most frequently reported measure of IT implementation success, we consider the implementation and use of the new change-request handling tool and process as indicative of the improvement success. This also aligns well to Seddon & Kiew's (1996) modification of DeLone and McLean's IT use to IT usefulness. Seddon and Kiew argue a non-mandatory usage of a system is a good proxy of usefulness and the new change-request tool was never discussed as mandatory to use (Brown *et al.*, 2002). The new change-request process was successfully diffused into the organization and the chasm was crossed. In the following, we describe how the change agents developed the guerilla tactic based on Moore's bowling alley approach (1999), and we discuss practical lessons learned with respect to crossing the chasm and the core agility values expressed in the *Manifesto for Agile Software Development* (Beck *et al.*, 2001).

The argument is presented as follows. First, we present the theoretical background. We then present, analyze, and discuss data from the improvement initiative according to Susman & Evered's (1978) cyclical action research approach. Finally, we present our conclusions.

Theoretical context

This section presents the theoretical background for our research. First, we present change-agent tactics. Second, we review the technology adoption curve (Rogers, 2003), the chasm concept (Moore, 2002), and Moore's (1999)

bowling alley strategy for crossing the chasm. Finally, we discuss software agility (Beck *et al.*, 2001) in relation to improvement practices.

Change-agent tactics

To achieve successful improvement initiatives, Humphrey (1989) stresses the need for *change agents*, which he defines as 'dedicated resources' who are enthusiastic, technically and politically capable, and have management's confidence. These dedicated resources are needed to assure that practitioners overcome resistance and get adequate training and support, and that projects receive necessary consultation. Other literature on improving software organizations also recognizes the role of change agents in successful improvement initiatives. McFeeley (1996) stresses the importance of committing dedicated resources to drive the improvement work. Fowler & Levine (1993) identify the change agents' role and tactics as one of five key factors for successful innovation diffusion. Weinberg (1997) calls change agents 'change artists' claiming 'change artists are the lubricant that makes change work' (Weinberg, 1997, p. 55). For change to occur, Kautz *et al.* (2001) argue change agents must use of a variety of tactics. Kautz *et al.* have therefore developed four change agent roles – technical expert, facilitating participant, political agent, and individual therapist – that change agents can use to understand more about the change-agent role and thereby contribute more to successful change.

Humphrey (1989), McFeeley (1996), Grady (1997), and Weinberg (1997) all agree that improvement initiatives will not happen (i.e., diffusion will not occur) if dedicated people do not drive the change (i.e., the innovation). This is also further elaborated on and emphasized by Börjesson & Mathiassen (2004), who point out several different change-agent tactics that help improvement initiatives in software organizations succeed. Rogers (2003) claims change agents would not be needed to diffuse innovations if there were no social and technical chasms between the change agency and the client system. This chasm exists, however, and Rogers therefore defines five different generalizations – change-agent tactics – for how change agents can interact with the clients to get positive diffusion effects. According to Rogers, change agents' ability to get clients to adopt new processes is positively related to:

- the extent of the change agents' effort in contacting clients;
- the change agents' having a client orientation (rather than a change-agency orientation);
- the compatibility between the diffusion program and the clients' needs;
- the change agents' empathy with clients; and
- the change agents' credibility in the clients' eyes.

Rogers (2003) also argues for the use of opinion leaders in securing innovation adoption. Rogers defines opinion leaders as individuals who provide information and

advice about the innovation to many other individuals. A change agent is preferably also an opinion leader, but an opinion leader must not be the change agent driving the change initiative. The opinion leaders are distinguished by high technical competence, high social accessibility, and high conformity to the organization's norms. The latter is particularly important, as individuals perceived as deviants are accorded low credibility. Similar to opinion leaders, Humphrey's (1989) champions are individuals involved in improvement initiatives. They maintain a focus on the goal, strive to overcome obstacles, and refuse to give up when the going gets rough. Humphrey claims that new ideas without champions have little chance of success. 'Although champions don't always win, winners, it seems, always have champions' (Humphrey, 1989, p. 31). Opinion leaders and champions differ somewhat in definition. Opinion leaders are individuals whom change agents use to assure successful change, while champions might very well be the change agents themselves. Pries-Heje (2003) defines a champion as the person who, in practice, affects the target user and ensures that the change occurs. This person could, according to Rogers' (2003) and Humphrey's (1989) definitions, be both an opinion leader and a change agent. In our study, we use the terms 'change agents' and 'opinion leaders' as defined by Humphrey (1989) and Rogers (2003), respectively.

Crossing the chasm

It is well established within the diffusion of innovation literature that there is a gap between acquired and deployed technologies (Fichman & Kemerer, 1999). Researchers have presented several theories to explain the reasons for this gap (Weinberg, 1997; Abrahamsson, 2001; Börjesson and Mathiassen, 2004), including Moore's chasm (2002). The chasm framework was originally developed to explain how high-tech products were marketed and deployed within a group of profitable firms. The chasm has also been discussed and used as an overall framework for understanding change (Paulk, 1999). Paulk argues that the use of structured tactics to cope with change increases the likelihood of successful change. Moore's suggestion for how to cross the chasm – the 'down the bowling alley' approach (1999) – is one such tactic. Even though the chasm approach was developed in a different context, we were inspired both by Paulk (1999) and Rogers (2003) to use it in our study. Rogers does not necessarily recognize the chasm as defined by Moore (2002), but he does recognize the need to deal with uncertainty and lack of structure. Paulk (1999) provided us with ideas about how to deal with uncertainty and lack of structure – that is, use structured change models. The chasm is one such approach suggested by Paulk.

Moore's chasm is built upon Rogers' well-known technology adoption curve (2003), which consists of five groups of people, each with its own specific profile (see Figure 1). The first group, *innovators*, tends to seek out the

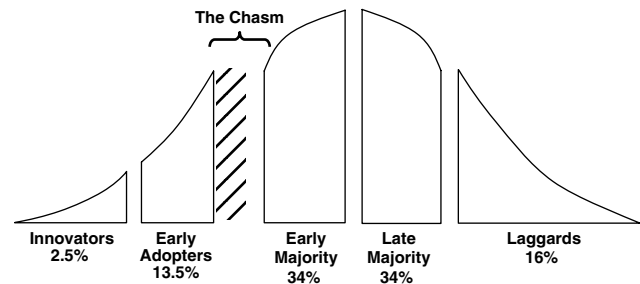


Figure 1 The technology adoption curve (Rogers, 2003) and the chasm (Moore, 2002).

new technology before it is formally marketed. Innovators are typically very technology focused. The second group, *early adopters*, accepts new products very early in the technology life cycle. Early adopters find it easy to imagine, understand, and appreciate the new technology's potential benefits. The third group, *early majority*, is driven by a strong sense of practicality. People in the early majority want well-established references from relevant market segments before committing themselves. The fourth group, *late majority*, shares all of the early majority's concerns and feels uncomfortable in their ability to manage the new technology. They therefore wait until a new technology has become a well-established standard. The fifth and last group, *laggards*, simply want nothing to do with the new technology. When they do accept a new technology, they typically do so without realizing it.

When diffusing an innovation, it is common to create an effect where each convinced group is used as a reference for the next group to keep momentum for marketing the new technology (Rogers, 2003). Between all such groups, there is a gap. This gap symbolizes the difficulty any group will have in accepting a new technology if it is presented to them in the same way it was presented to the previous adopters (Moore, 2002). As Figure 1 shows, the key gap is the chasm between early adopters and the early majority (Moore, 2002). Moore claims that this chasm is extremely dangerous if it goes unrecognized. The innovators and early adopters will bear with bugs and glitches for the sake of a new technology. The early majority wants technology to enhance, not overthrow, established ways of working. Given these incompatibilities, early adopters do not make a good reference for the early majority. Nonetheless, the early majority needs good references that assure a minimal disruption of their established way of working. Moore (2002) defines this as a catch-22.

To solve the catch-22, Moore (1999) presents a tactic for crossing the chasm called 'down the bowling alley' in which each market niche (or group of would-be adopters) is represented as a bowling pin. The goal is to strike down one pin so that it helps 'knock the next one over'. In this way, momentum is built up as more and more pins are knocked over. Moore claims that selecting the pins to

target is an important task. The initial pins should meet two criteria: the would-be adopters should have a compelling need for the solution, and competition should be scarce in that area. The innovation suppliers should also apply a customer-oriented tactic, ensuring that they can deliver the whole product to the target adopters in a way that meets their particular expectations (vs those of the whole market). Moore (1999) identifies the 'whole product' as the bare minimum requirements that must be met for a specific customer to purchase the innovation. Momentum is built up through first selecting pins from within a single market segment with the same whole product needs, and then further developing the product to reach new market segments. At this point, the chasm has been crossed, and the early majority market is about to be penetrated. Table 1 below describes the main elements in the chasm concept and the 'down the bowling alley' tactic and how these elements match up with our study's context. The similarities between the chasm concept and the studied context make the target improvement initiative prone to fall into the chasm.

Agile improvement practices

Agility implies speed, although being fast does not necessarily imply being agile (Levine, 2005). Agility is of rather little interest when environmental changes are slow and predictable (Haeckel, 1995). In an ever-changing business environment, however, organizations must develop a conscious agility competency to survive (Dove, 2001). The ability to respond to environmental events is hence the essential and distinguishing feature of the agile organization (Haeckel, 1995, 1999; Dove, 2001).

There are currently numerous definitions for agility. Because our research addresses improvement practices in software organizations, we are partial to the *Manifesto for Agile Software Development* (Beck *et al.*, 2001). Agile alliance members have expressed four preferences and values:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

The *Manifesto* says that while the items on the right side of these preferences have value, those on the left have a higher value. There is an ongoing debate on how to interpret the *Manifesto* values. Levine (2005) argues that customers' and developers' interests in agile practices are necessarily different. Development efforts embracing the *Manifesto's* left-side values are often at a loss when it comes to governance and monitoring. Also, Rakitin (2001) claims that the left-side values serve as an excuse for developers to throw code together with no regard for engineering disciplines. Rakitin suggests that some developers might translate the agile *Manifesto* as: 'We know what you want – even if you don't', 'Real programmers don't write documentation', 'Let's not spend time haggling over details', and 'Why spend time on planning when we could be coding'. Levine (2005) argues that the software industry faces a challenge when trying to interpret and successfully implement agile values.

At this point, understanding about how agility impacts improvement practices is limited. Holmberg & Mathiassen

Table 1 Similarities between the Ericsson context and the main elements in the chasm concept and 'down the bowling alley' tactic

<i>The chasm/down the bowling alley</i>	<i>The studied context</i>
Deal with high technology products that are complex and multifaceted	The studied change-request process and tool manages thousands of change requests. The change requests are read by hundreds of individuals and managed by several steering groups. Because many different individuals are involved with and using the solution, dealing with all their opinions, requests, and needs is complex
Niche of potential users	The Ericsson company consists of different categories of people with different requirements for the change request process and tool. The different groups' requests differ depending on their responsibility (investigate, prioritize, decide, administer, or design a solution based on the change request)
Whole product	The innovation suppliers, that is, change agents in this study, must deliver functions based on the category of people who require prioritized support. Given reactions to change (Weinberg, 1997) and the burden of learning (Attewell, 1992), it's too hard to provide all functions to all categories in one delivery
Bowling-pin strategy	When the basic functions in the change-request process and tool are delivered and used by well-known and accepted colleagues, more people are likely to use them, especially if the process and tool are complemented with functions focusing on the target group's tasks and responsibilities
Profound difference between the innovators and the early majority	At Ericsson, some employees love the technical challenge of a new process and tool, can bear with bugs, and can easily imagine all potential benefits. Other employees only want a new technique if it is stable and offers proven benefits to their daily work. Our experience and extensive participation in previous improvement work taught us that there is a profound difference in employees' attitudes toward new techniques, and that not considering this difference might lead to change failure

(2001) explain how improvement efforts are sidetracked and interrupted when organizational changes occur. Börjesson & Mathiassen (2005) provide similar explanations about improvement practices and organizational dynamics, cautioning that agile improvement practices require coordination and alignment with other initiatives to increase the software organization's ability to respond. For software improvement, Börjesson & Mathiassen (2005) highlight particular agility challenges, including the need to sense and respond to changes in software development, to sense and coordinate with other change initiatives within the organization, and to provide a flexible process infrastructure. There is, however, little knowledge about how organizations can adopt agile improvement practices to enhance diffusion success.

Research approach

Our improvement initiative had two goals: to improve Ericsson's change-request handling and to contribute to the literature by addressing uncertainty through agile improvement practices in software organizations. Mathiassen's collaborative practice research (CPR) supported the possible realization of this dual goal (2002), as an important aspect of CPR is the emphasis on insider and outsider perspectives (Bartunek & Louis, 1996). In our case, all three of us were working within Ericsson's software improvement unit, but two of us were driving the initiative as change agents, taking insider roles and being directly involved in action. The third author was responsible for the initiative, but assumed the outsider role through regular analysis, discussions with the change agents, and reflections and interviews with participants in the change-request initiative. The initiative was also discussed, analyzed, and reflected upon in a monthly software improvement forum (Börjesson, 2006).

Our study was based on action research (Galliers, 1992; Baskerville & Pries-Heje, 1999; Davison *et al.*, 2004) with a focus on understanding how change agents can adopt agile improvement practice to cross the chasm and by doing so assure implementation success. Our research question was: *'How can change agents in software organiza-*

tions adopt agile improvement practices to successfully cross the chasm?' Baskerville & Pries-Heje (1999) argue that the fundamental contention of action research is that a complex social process can be studied best by introducing changes into that process and observing the effects of these changes. Recent research by Davison *et al.* (2004) elicits a set of five principles and associated criteria to ensure and assess both the rigor and relevance of canonical action research (CAR). The term 'canonical' is used to formalize the association with the iterative, rigorous, and collaborative process-oriented model developed by Susman & Evered (1978). Susman and Evered's model has been widely adopted in the social sciences and hence gained 'canonical status' (Davison *et al.*, 2004). Table 2 shows the five proposed principles and how we adopted them in this study.

Both the study and the structure of this paper follow Susman & Evered's (1978) cyclical action research approach and its five phases: diagnosing (identify the problem), action planning (consider alternative courses of action for solving the problems), action taking (select a course of action), evaluating (study the consequences of the actions taken), and specifying learning (identify general indicators for implementation success).

We collected data throughout the 16-month initiative, as summarized in Table 3. Our data analysis was based on triangulation (Yin, 1994), that is, combining different data sources to avoid bias and secure the validity of the research.

Action research

Both our improvement initiative and the following discussion follow Susman & Evered's (1978) five phases. The first section on Diagnosing identifies and defines the problem with the existing procedures for change-request handling and the difficulties with assuring implementation and use. The second section on Action Planning describes the plan to solve the problems by implementing a new solution for change requests; it also describes a new tactic to address the difficulties with process implementation and use. The third section on Action Taking describes which actions were taken according to the

Table 2 Canonical action research principles (Davison *et al.*, 2004) and how they were adopted

Principle	Adoption in this study
Researcher–client agreement (RCA)	The researchers and clients were the same individuals. One of the three researchers assumed the academic role through her commitment to an ongoing industrial Ph.D. program
Cyclical process model (CPM)	We used Susman & Evered's (1978) cyclical action research method
Theory principle	We used four established concepts: the technology adoption curve (Rogers, 2003), the chasm (Moore, 2002), 'down the bowling alley' (Moore, 1999), and the <i>Manifesto for Agile Software Development</i> (Beck <i>et al.</i> , 2001)
Change through action	One of the study's main goals was to change the current way of working and improve software practice. We adapted CPR (Mathiassen, 2002)
Learning through reflection	We planned and held regular meetings for reflections on the initiative's ongoing progress. We regularly analyzed and reflected upon the initiative in a monthly improvement forum (Börjesson, 2006)

Table 3 Data collected throughout the initiative

No.	What	Explanation
1	Direct involvement	All three authors were directly involved in or responsible for the improvement initiative's management and outcome. This participation gave them direct access to the organization, personal opinions, coffee break discussions, etc., which would not have otherwise been possible
2	Improvement initiative data	The two change agents driving the initiative collected data about reference group meetings (time, participants, disagreements, decisions, outcome, etc.)
3	Participatory observations	The author taking the outsider role participated in some reference group meetings to observe and reflect
4	Software Improvement forum content	We studied, discussed, and reflected upon various papers and ongoing improvement initiatives
5	Software Improvement forum reflections	Comments, ideas, and reflections during the improvement forum were written down and stored by one of the authors on the internal unit website
6	Software Improvement survey 2003 and 2004	We asked all development unit software engineers 60 questions (with six-scale answers) through a Web questionnaire in the annual software improvement survey
7	Tool data	We had access to tool data, including change-request registrations, who and how many people had entered data, and how change requests were managed according to the process
8	Qualitative interviews with eight reference group participants	Each interview lasted for at least 30 min; two of the authors were present at each interview

Table 4 The change-request process and tool situation in late 2003

CCB	Process	Tool
Development project A	Project-specific process A	No tool (spreadsheet A)
Development project B	Project-specific process B	No tool (spreadsheet B)
Development project C	Project-specific process C	Local version C
Unit X	Unit-specific process X	Local version X
Unit Y	Unit-specific process Y	No tool (spreadsheet Y)

plan. The fourth section on Evaluating discusses the evaluation of the actions taken. Finally, section five on Learning discusses general indicators for successful change in relation to the theoretical context discussed in our Action Planning section.

Diagnosing

In late 2003, the change-request procedures were identified as a major risk for development projects and the organization as a whole (see Table 4). Not only was it hard to get an overview of change requests, the existing change-request procedures were inadequate. A typical change-request procedure includes activities and support for writing, submitting, analyzing, assigning, deciding, and validating change requests. In recent years, the increasing pace of environmental change has given rise to the notions of fast-moving software organizations (Baskerville *et al.*, 2001; Holmberg & Mathiassen, 2001; Börjesson & Mathiassen, 2005) and radical IT-based innovations (Lyytinen & Rose, 2003). As a result of the changing business environment, software organizations and their ongoing development projects must constantly react to new or changing customer requirements. At Ericsson, change control boards (CCBs) handle change requests in development projects or specific organizational units. When we started our initiative, the CCBs

were all using different processes for change-request handling, and most processes lacked tool support. There was a high degree of uncertainty involved in the change-request procedures. Table 4 shows the change-request process and tool situation in late 2003.

Whenever a project was defined, a CCB was set up to support that project. Because there was no common change-request process or tool, much effort was spent on defining these procedures for each project. As a result, each CCB had its own process for change-request handling, and it was complicated to manage change requests that involved more than one CCB. The lack of common CCB procedures increased both administration of change requests and the risk of losing track of decisions and information between CCBs.

In addition to this complex change-request situation, we had identified a problem with deploying new solutions successfully. Through extensive improvement initiative participation, theory studies, discussions, and reflections (Börjesson, 2006), we realized that the improvement unit faced major difficulties when trying to widely deploy a new or changed process (Humphrey, 1989; Weinberg, 1997; Fichman & Kemerer, 1999; Börjesson & Mathiassen, 2004). In general, users did not accept new solutions as easily as the change agents driving the initiatives. Many potential users were driven

by a strong sense of practicality, and they wanted clear proof that the new solution would benefit them. Data from two consecutive software improvement surveys showed that people wanted to be involved and participate in defining the solution to assure successful change (see Table 5). The surveys were sent to all 500 employees, and 200–220 of them responded each year. Among the findings was that almost three out of ten employees wanted to lead the work; more than six out of ten employees wanted to help define the new way of working; and more than eight out of ten wanted to review the new approach to assure it would be good enough. It is, however, practically impossible to fulfill these needs in a large organization. As this data indicates, it is important to have trustworthy references that people can rely on.

Action planning

In late 2003, the organization decided to start an improvement initiative to address the uncertainty and improve the change-request process and tool situation. The plan was to define a new process (based on previous experiences) in close collaboration with a recently started development project and the two units managing overall product requirements. CCBs were then to align with the new process as new development projects started. For our tool, we selected ClearQuest, which Ericsson's corporate headquarters had decided to use for change-request handling. Also, two CCBs had used ClearQuest before and it was connected to other tools in use at Ericsson, including RequisitePro and ClearCase.

In parallel with this effort, the improvement unit decided to use previous learning on how to manage successful improvement initiatives within the organization (Börjesson & Mathiassen, 2004). This learning showed that dedicated improvement initiatives were more successful than generic ones. However, the whole organization needed a common change-request solution. We therefore decided to use a reference group that could become the dedicated receiver of the change-request solution. Previous learning also showed that a successful improvement initiative needed both practice pull and process push. This learning had been documented in a process to support the change agents' daily work. We therefore planned to have the two change agents driving the initiative facilitate process push, and the reference group facilitate practice pull. Finally, throughout the initiative, the two change agents would consider previous

learning regarding the need to focus on deployment activities and perform iterative improvement work.

The two change agents also decided to try out a tactic aimed at getting new users to accept the solution faster. Because the identified problems related to the chasm (Moore, 2002), they used Moore's bowling alley approach (1999) to set up the guerrilla tactic. The agile *Manifesto* (Beck *et al.*, 2001) inspired the change agents' thinking and acting. This was particularly true in their interactions with the reference group to create a working solution and in their ability to reflect on and respond to that group's changing requirements. Also, the change agents wanted to address the uncertainties involved in the improvement initiative through a series of interactions between themselves and the adopters. The guerrilla tactic consists of two main parts: choosing a dedicated reference group with a compelling need for the solution, and having a tactic for close collaboration between that group and the change agents. Having a reference group that all potential users could rely on is a benefit for improvement initiatives. The reference group should consist of highly practical people who can serve as a natural reference for the majority of users. In our case, the reference group consisted of members from all affected organizational areas. Generally, reference group members must be experienced in the specific practice, understand the big picture, and be well known and respected among their peers to facilitate deployment of the improvement initiative. Change agents are the people who easily understand and appreciate the new practice's potential benefits; as such, they should collaborate closely with the reference group to become their natural reference. The change agents should also strive to influence strategic individuals within the reference group. The goals here are to facilitate commitment at reference group decision meetings and to gain new knowledge by exploring ideas with individual group members.

The collaboration between the reference group and the change agents can be described as 'influencing a group by influencing its core members'. To achieve this, our change agents personally briefed and discussed upcoming decisions with key reference group members in between reference group decision meetings. The members who needed to be addressed before a decision meeting typically varied, depending on decision areas and the group members' individual understandings and beliefs. The change agents carefully observed the situation at each reference group meeting and responded

Table 5 Data from software improvement unit survey in 2003 and 2004

How important is it that you participate to assure successful change?	Not important (%)		Important (%)	
	2003	2004	2003	2004
I am the leader of the process work	72	71	28	29
I am a part of the group defining the new way of working	40	38	60	62
I give feedback on the new way of working to ensure that it is good enough	16	14	84	86

accordingly. Collaboration with strategic reference group members should address problems and uncertainties, as well as new ideas for how things might be solved before the actual decision meetings take place.

The guerrilla tactic reflects Moore's (1999) bowling-pin thinking. The reference group represents the first bowling pin and helps knock other pins over (i.e., they get new people to use the solution; see Figure 2). In our case, the second-row bowling pins were CCB participants, project managers, and change-request administrators. In the third row were people who handled a lot of change requests within the projects. That row was followed by rows of different users, such as testers and programmers. It is important to choose and collaborate with reference group members in a way that ensures their commitment and dedication to the solution. When planned properly, reference group members can serve as references for all other potential users.

Action taking

Our project focused on assuring implementation and use of the new change-request solution throughout the initiative. Our first action was to set up the reference group. The change agents asked nine individuals responsible for or involved in the CCB work to participate in the reference group. These individuals had high conformity with the organization, as well as excellent technical competence and social skills. The reference group members should function as opinion leaders (Rogers, 2003) for the change-request solution across the organi-

zation. It is therefore important to find reference group members with attributes that facilitate this role.

As they formed the group, the change agents simultaneously started building their own competence. To be able to judge the validity of different reference group proposals, the change agents had to master both the change-request process area and the ClearQuest tool. One important success factor for high process push is to have change agents who are both committed *and* competent (Humphrey, 1989; McFeeley, 1996; Börjesson, 2006).

Between reference group meetings, the change agents addressed strategic group members to prevent unnecessary disagreements and improve the solution. The change agents might seek out a person if they thought, for example, that he or she had in-depth knowledge, was poorly informed, felt overwhelmed, or could help other group members understand the big picture. The change agents focused on getting the tool solution developed as decisions were made in reference group meetings. At the end of the improvement initiative, the change agents documented critical parts of the tool to assure its future maintenance. They also documented the change-request process in parallel with the reference group meetings to provide discussion material and to assure that everyone understood the agreements. The reference group had six meetings; in four of them, all necessary decisions and agreements were reached. These four meetings were preceded by individual collaboration with strategically chosen reference group members. The reference group did have disagreements, however, and consensus

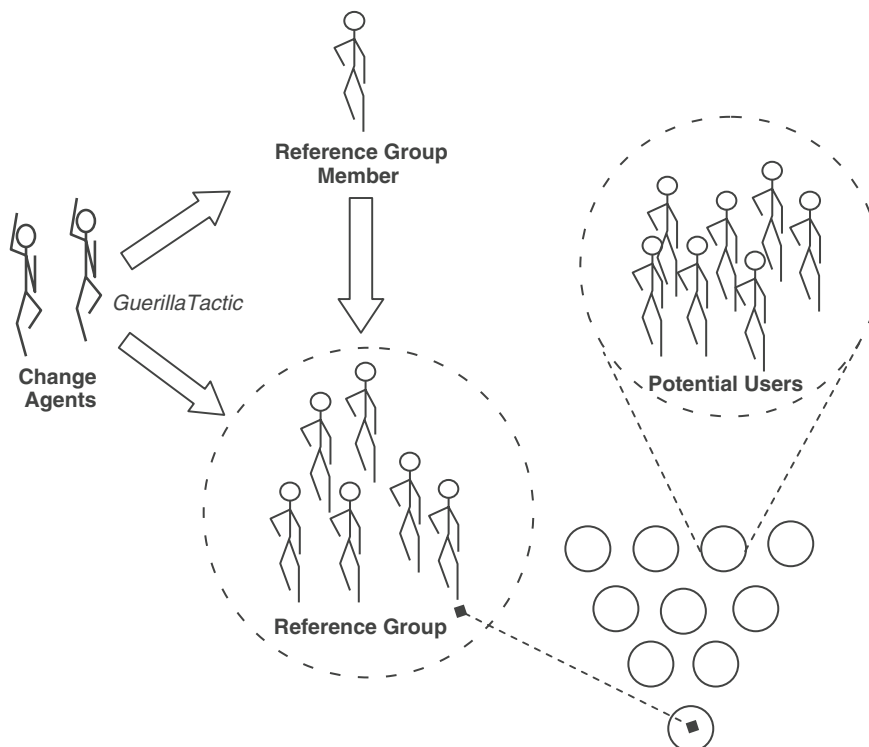


Figure 2 Change agents using the 'guerrilla tactic'.

decisions were not always possible. The two change agents focused on the big picture and helping the reference group understand the main problem. This facilitated decision-making, which was a prerequisite to meeting the improvement initiative's proposed schedule. The two meetings not preceded by individual collaborations failed to produce necessary decisions on how to proceed with the solution.

In early May 2004, the new change-request solution was presented and deployed. The change agents gave several courses to all potential users. Reference group members participated in each course. Because they were active users of the new solution, the reference group members could defend and explain it, and became well-established references. The change agents also gave start-up meetings for the affected CCBs to support them as they introduced the new way of working. Again, reference group members played a central diffusion role, since they were both regular CCB members and a part of the improvement initiative. Finally, as the change agents were phased out from deployment activities, the reference group members stayed current in their traditional CCB roles and supported the new solution's day-to-day diffusion and use.

Evaluating

The reference group meetings varied in success depending on whether or not the change agents collaborated with group members between meetings. Table 6 shows the date, purpose, and pre-meeting collaborations for each reference group meeting, as well as the meeting's outcome. Those meetings prepared for according to the

guerrilla tactic were successful, whereas other meetings were less successful.

In mid-June 2004, we interviewed eight of the nine reference group members (one being on vacation) regarding their experiences with and opinions about the improvement initiative and solution. Our goal was to further understand how the guerrilla tactic had affected the initiative's outcome. At the time of the interviews, reference group members were unaware of the guerrilla tactic. The interviewees represented the targeted CCBs, CCB chairpersons, project managers, and senior engineers working with change requests in their daily work. Table 7 summarizes the interview results.

Negative answers in the interviews revealed that some people were not as happy with the outcome as others. Those negative answers were mainly given by the same two people. One of those individuals needed features in the new solution that were given low priority by the reference group in order to get a first release ready according to plan. Although these features were not needed to start using the new procedures, they were needed to facilitate change-request administration at the CCBs. The other person who responded negatively was generally unable to see benefits from the new solution in his daily work.

When the solution was released, we gave all potential users immediate access to the tool. Table 8 shows the growth of users who have entered data in the tool and the amount of data in the database. The table's numbers accumulate from the solution's release date (May 10, 2004) and show the number of individuals who updated

Table 6 Reference group meeting data

Reference group meeting	Pre-meeting collaboration	Meeting outcome
No. 1: 2004-01-15 Present proposed schedule, purpose of the reference group, and issues related to change-request handling	The change agents collaborated individually with four reference group members	Agreement; necessary decisions made
No. 2: 2004-01-30 Present overall change-request solution and concept	No collaboration	No agreement. Most reference group members had differing opinions about a potential solution
No. 3: 2004-02-06 Present overall change-request solution and concept	The change agents collaborated individually with five reference group members	Agreement; necessary decisions made
No. 4: 2004-02-20 Present change-request process and detailed solution	The change agents collaborated individually with three reference group members	Agreement (after minor adjustments); necessary decisions made
No. 5: 2004-03-05 Peer review of detailed solution; two new reference group participants added	No collaboration	No agreement. The two new reference group members objected, as they did not fully understand the earlier agreements
No. 6: 2004-03-12 Formal review of detailed solution and peer review of tool interface	The change agents collaborated individually with seven reference group members	Agreement (after minor adjustments); necessary decisions made

Table 7 Compiled interview results

No.	Compiled answers	Yes	No	Not applicable	Positive answers (%)
1	The result was generally positive	6	2	0	75
2	The decisions were generally handled in a positive way	6	1	1	75
3	The change agents' process push was generally positive.	5	1	2	63
4	The rollout was generally considered as positive	7	0	1	88
5	The users' responses have generally been positive	6	2	0	75
6	The attitude is, in the end, generally positive	6	2	0	75
7	It was generally positive with a tight time schedule	6	1	1	75
8	The result was generally received positively	5	1	2	63

Table 8 Active users and data in the change-request database

Measurement	Cumulative number of submitted change requests	Cumulative number of unique users who had updated change requests
May 2004	24	19
June	99	42
July	170	52
August	258	52
September	342	63
October	449	73
November	548	85
December	623	91
January 2005	689	99
February	782	112
March	866	126
April	958	139

information in the tool. We were unable to measure users who simply read information.

We carried out the change-request initiative according to the original proposed schedule. The six reference group meetings were held within the planned time frame. We originally planned to rollout the solution in mid-April, to match the needs of a new engineering project. That project was a few weeks late, however, so we moved the rollout of the new change-request solution to early May.

Our improvement initiative aimed to cover the whole organization, but not all employees are required to work with change-request handling. We calculated the total target group based on the main technical stakeholders (in all, 115 CCB administrators, project managers, and requirements engineers) having an 80% usage, and peripheral technical stakeholders (398 design and test engineers) having a 25% usage. We base the 80% estimate on the fact that most, but not all, main technical stakeholders work with change requests, while the 25% estimate is based on the fact that only team leaders and senior engineers work with change requests. We did not include groups that are not expected to work with change-request handling (43 employees in all). The authors have total access to employee databases to

Table 9 Distribution of potential users between groups

Group	Percentage according to Rogers (%)	$192 \times \text{Rogers' percentage}$
Innovators	2.5	5
Early adopters	13.5	26
Early majority	34	65
Majority	34	65
Laggards	16	31
Total	100	192 ($115 \times 0.8 + 398 \times 0.25$)

validate the number of employees and their roles. This, combined with extensive participation in the change-request practice, made it possible to calculate the expected number of change-request users. Table 9 shows the population distribution of expected change-request users and the role they play in Rogers' diffusion of innovation curve (2003).

Learning

This section discusses the learning from this study. The first section presents how the new process and tool were diffused in the organization and discusses it in relation to Rogers' (2003) technology adoption curve and Moore's chasm (2002). The second section discusses the impact the guerrilla tactic had on the improvement initiative's outcome and relates the tactic to Moore's (2002) bowling alley analogy. The third section discusses learning from the guerrilla tactic in relation to the agile *Manifesto* (Beck *et al.*, 2001). The fourth and final section discusses possible limitations with the study.

Crossing the chasm As indicated in the interviews (Table 7) and tool measurements (Table 8), the improvement initiative was successful in both implementation and use. Seventy-five percent (six of eight individuals) of the interviewees viewed the initiative as successful. Table 8 shows a regular growth of ClearQuest tool users since its release. At the time of the interviews – 2 months after launch – we were achieving initial penetration in the early majority. In theory, we had crossed the chasm,

which required 31 individuals to cross the early adopters (see Table 9). At that time, we had 42 unique tool users (see Table 8). By September, 5 months after rollout, half of the early majority had been captured. This data indicates that, in practice, we had crossed the chasm: 63 unique users were now using the tool. To capture the whole group, up to and including early majority, we needed 96 individuals, according to Rogers' percentages (2003). We achieved this in January 2005, 9 months after rollout. Twelve months after rollout, more than half of the late majority had been captured: 139 out of 192 estimated unique users were now using the tool. Figure 3 shows the accumulated number of unique tool users over time and the improvement initiative's main events. The data shown in Figure 3 strongly indicates both that the chasm was successfully crossed and that most possible users were active in the ClearQuest tool. From the interviews, we learned that there was a high adherence to the process. The initiative had been successful in diffusing the new change-request process and tool.

The main issue now is to understand what effects the guerrilla tactic had on this positive outcome.

The Guerrilla tactic The two change agents did not believe they could determine the problems and solutions for change-request handling on their own. They therefore used tactics that facilitated close collaboration with strategic users of the new process. They also tried to solve conflicts, not only as they occurred, but also through preventive actions. During the process, structural

conflicts over proposed improvements and solutions arose among reference group members from different organizational areas. To deal with this, the change agents used tight and focused collaboration to influence group tensions and contradictions. Rogers (2003) identifies this collaborative tactic as one of five key features for change agents to succeed in securing innovation adoption.

In selecting the reference group, the two change agents identified individuals with attributes representing the early majority group. As such, these individuals could become natural references and act as opinion leaders (Rogers, 2003) to help the rest of the early majority cross the chasm (Moore, 2002). This tactic closely relates to selecting the first pin in the bowling alley approach (Moore, 1999). Identifying this first pin is crucial, according to Moore (1999), who claims that every other pin is 'derived' from this head pin. In our study, we realized that the reference group would be key to capturing new users. These reference group members established themselves as opinion leaders (Rogers, 2003) by defending and explaining the solution. Selecting reference group participants was thus essential for crossing the chasm; most users were 'derived' from the reference group.

Most reference groups have members who are initially protective of the existing way of working and thus require special treatment. In our case, the change agents used collaboration tactics – both one-on-one and in the group – to give each reference group member special attention and manage their individual needs. This

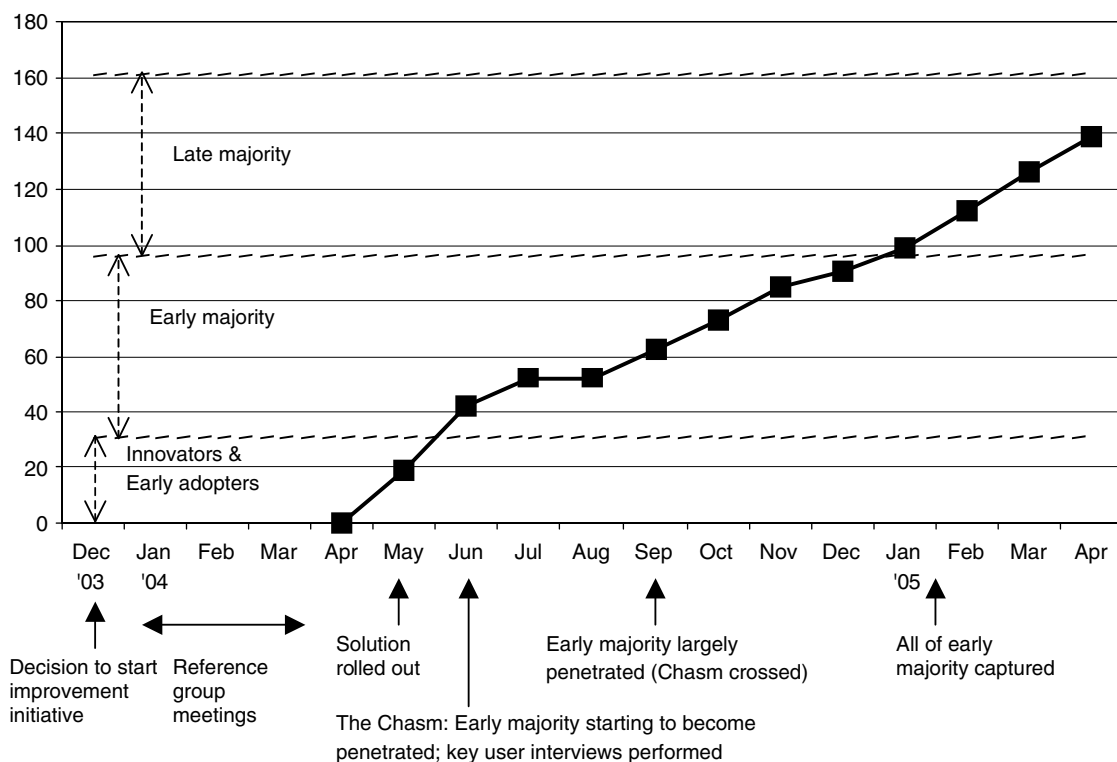


Figure 3 Main events and increase of unique users updating change requests in the tool.

supports Moore's ideas about building momentum so that one pin can help knock over the next one (1999) (see Figure 2). Furthermore, the change agents listened to and collaborated closely with reference group members to understand and identify key features that would help users embrace the solution. This part of the tactic is well aligned with Moore's identification of 'the whole product' (1999). The guerilla tactic is partly an example of the bowling alley approach. Figure 4 explains how the bowling alley and guerrilla tactics relate to each other.

As data from Table 6 shows, there is a strong correlation between high use of the guerrilla tactic and high acceptance of the solution suggested by the change agents. The reference group meetings that were prepared for through such individual collaboration resulted in a more positive result than those that occurred without this collaboration. In the successful decision meetings, the reference group members had a tendency to focus and agree on important issues that let us move forward.

User response to the new solution has been generally positive (see Table 7). We encountered no particular turmoil that required extra and unplanned efforts to manage reactions to change or capture the early majority group. Because reference group members work in the daily business where the new solution is being used, they have been able to function as natural references for the rest of the early majority group. Not one project or line manager expressed dissatisfaction with the solution during rollout. The chasm was successfully crossed and the new solution was implemented and used, indicating improvement success (DeLone & McLean, 1992). Our study indicates that the guerrilla tactic had a positive impact on the outcome of the improvement initiative –

that is, that there are specific tactics, such as the guerrilla tactic, that change agents can use to cross the chasm.

Agile improvement practices Levine (2005) has identified a challenge in further interpreting and implementing the agile *Manifesto* (Beck *et al.*, 2001) within software organizations. There is a delicate balance between the *Manifesto's* right- and left-side values, and neither is sufficient on its own. The *Manifesto* itself recognizes value on both sides, but states that it values items to the left (e.g., individuals and interaction) over items to the right (e.g., processes and tools). The *Manifesto's* use of the term 'over' has been interpreted in unintended ways, and its overall disclaimer – that both sets of values should be appreciated – seems to be ignored. As a consequence, the core left-side values are often used as an excuse to overrule the right-side values, as when software developers are encouraged to skip documentation and requirements understanding (Rakitin, 2001).

Using the guerrilla tactic can help practitioners interpret the agile *Manifesto's* (Beck *et al.*, 2001) core values for agile improvement practices. Table 10 illustrates guerrilla tactic features that are relevant to the *Manifesto*. We support the *Manifesto's* values. However, to avoid any misinterpretations, we argue for agile improvement practices as well as for attending to the core right-side values. We agree that, for instance, 'individuals and interactions' is more important than 'processes and tools', but this must not be interpreted as 'no processes and tools are needed at all'. As the Table 10 below shows, change agents in our study combined values from both the left and right sides.

	"Down the bowling alley" (Moore, 2002) (Each market niche, or group of would-be-adopters, is represented as a bowling pin; the goal is to strike down one pin so that it helps to knock over the next one)	Guerrilla tactic (Identify and collaborate with respected practitioners who can understand and appreciate the new practice and become opinion leaders and natural references for the majority of users)
How to choose first target group	Choose an initial "pin" with a compelling need for the solution. Assure scarce competition in the selected pin area.	Choose a dedicated reference group whose members can serve as natural references for most potential users. Choose group members from all affected organizational areas. Group members should be well known and respected, be capable of serving as opinion leaders, be experienced in the change area, and understand the big picture.
How to meet customer requirements	The innovation supplier must ensure that they can deliver the whole product to the selected target (that is, rather than trying to satisfy the whole market's requirements, the focus is on meeting one target customer's expectations). Moore (1999) identifies "the whole product" as the bare minimum requirements that must be met for a specific customer to purchase the innovation.	The change agents give lower priority to features that are not required to get a first release rolling as scheduled. The change agents collaborate closely with the reference group to become their natural reference. Through this close collaboration, the change agents understand and identify the key features for the users. The change agents influence the most strategic individuals in the reference group in order to get commitment at reference group decision meetings.

Figure 4 The relations between 'down the bowling alley' and the guerrilla tactic.

Table 10 The agile *Manifesto* and the guerrilla tactic

<i>Agile manifesto</i>	<i>The guerrilla tactic</i>
Individuals and interactions over processes and tools	Close collaboration between change agents and reference group members based on previous learning (Börjesson & Mathiassen, 2004)
Working software over comprehensive documentation	Change agents focused on implementation and use of the new solution. Process description was made as a parallel activity to assure implementation
Customer collaboration over contract negotiation	Change agents and reference group members interacted on a daily basis. Agreements were made at reference group meetings
Responding to change over following a plan	To meet the project plan, the change agents had to continuously sense and respond to unplanned events and reactions from reference group members

According to Börjesson & Mathiassen (2005), the challenges for software improvement are to sense and respond to changes in software development, sense and coordinate with other change initiatives within the organization, and provide a flexible process infrastructure. The guerrilla tactic helps change-agents sense and to respond to situations by encouraging their strong collaboration with reference group members. This way of working is also supported by Rogers (2003), who argues that change agents' success in securing the adoption of processes by clients is positively related to the extent of the change agent's effort in contacting clients. We argue that the guerrilla tactic supports agile improvement practices in that it addresses uncertainty, promotes interaction, and facilitates a positive response to change. Therefore, the guerrilla tactic helps change agents cross the chasm, leading to successful process implementation. The agile *Manifesto* clearly states, 'while there is value in the items on the right, we value the items on the left more'. For agile improvement practices, we suggest that this statement be directly integrated in the four core values to avoid immature misinterpretations.

Future research The chasm concept was originally developed in a context other than software improvement environments. However, the many similarities we found between key elements in the chasm concept and in Ericsson AB's software improvement environment inspired us to use this concept. Several factors aside from the guerrilla tactic could have had a positive influence on our improvement initiative's outcome. Experienced change agents (Humphrey, 1989) with deep technical skills in the improvement area, the use of a tool to enforce the improvement (i.e. there is no way to bypass the process, since the tool handles all the information) and the organization's desire for a common change-request solution indicating high practice pull from the practitioners (Zmud, 1984; Börjesson & Mathiassen, 2004) are all factors that could have impacted the outcome. These issues are not sufficiently discussed in

this paper and further research is needed to identify the impact of these factors. We encourage practitioners and researchers to further experiment with and study how change agents can adopt agile improvement practices to cross the chasm. We need to understand and learn more about the chasm's perils and challenges for agile improvement practices to ensure successful implementation and use of new innovations.

Conclusions

Inspired by the chasm concept and its similarities with an upcoming improvement initiative, three change agents developed an agile improvement practice to help their software organization accomplish successful implementation. They called this agile improvement practice 'the guerrilla tactic' and built it on Moore's (1999) 'down the bowling alley' approach. The improvement initiative aimed to diffuse a new change-request process and tool to a development unit within the telecom company Ericsson AB in Gothenburg, Sweden. The guerrilla tactic identifies how change agents can select a target group and meet its requirements, thus enabling a smooth chasm crossing. The guerrilla tactic is discussed in relation to diffusions of innovation and software agility literature. Data from the study shows that the chasm was crossed 5 months after rollout, and the late majority was heavily penetrated 1 year after the release. The study indicates that the guerrilla tactic had a positive impact on the outcome of the improvement initiative. The authors argue that the guerrilla tactic is an agile improvement practice and that its use facilitated successful implementation in the software organization. The authors also argue the guerrilla tactic can be used to make agile improvement happen. Although often misinterpreted, the agile *Manifesto* clearly states, 'while there is value in the items on the right, we value the items on the left more'. Our experience suggests that the first part of this statement requires greater emphasis than it typically gets if teams are to achieve agile – and successful – improvement practices.

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